

GRENONBLE LE VERSOUD AP, France

WMO: 074870

Lat: 45.2173N

Lon: 5.8487E

Elev: 722

StdP: 14.32

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	23.8	14.1	11.6	24.7	18.5	14.4	28.2	18.2	53.1	15.4	44.6	3.1	30	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	91.1	71.8	88.1	70.4	85.1	69.1	73.3	87.6	71.7	85.3	70.1	82.6	5.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.6	108.1	79.5	67.1	102.2	77.6	65.7	97.3	75.8	37.3	87.9	35.8	85.6	34.4	82.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	13.6	11.0	16.9	95.4	6.7	2.9	12.1	97.5	8.1	99.2	4.4	100.9	-0.5	103.0		
(5)			16.0	77.5	6.5	1.9	11.3	78.9	7.5	80.0	3.9	81.1	-0.8	82.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	37.2	39.4	47.6	55.4	60.9	68.8	71.4	70.0	63.8	56.1	45.1	37.7		
	DBStd	13.69	6.20	6.65	6.12	6.08	6.35	5.78	5.54	5.46	5.34	6.23	6.85	6.05		
	HDD50	1417	399	299	119	20	2	0	0	0	0	21	174	383		
	HDD65	4432	863	718	540	289	159	27	13	18	82	281	596	846		
	CDD50	3073	1	2	44	183	340	564	664	620	415	209	28	3		
	CDD65	611	0	0	0	2	32	141	212	172	48	4	0	0		
	CDH74	6054	0	0	2	108	403	1375	2093	1606	425	42	0	0		
	CDH80	2120	0	0	0	13	95	485	824	609	92	2	0	0		
	WSAvg	3.1	2.7	3.3	3.6	3.7	3.6	3.6	3.4	3.0	2.7	2.6	2.7	2.6		
	PrecAvg	47.6	3.8	3.2	3.2	3.8	4.3	3.8	3.6	3.7	4.4	4.6	5.0	3.9		
(15) Precipitation	PrecMax	58.9	10.5	9.8	11.2	8.5	8.3	6.7	8.7	6.6	10.9	12.0	12.9	9.3		
	PrecMin	35.8	0.6	0.4	0.7	1.1	1.7	1.4	1.1	0.8	0.8	0.7	1.2	0.1		
	PrecStd	6.8	2.2	2.2	2.1	1.8	1.7	1.5	1.8	1.3	2.5	2.3	2.9	2.1		
	PrecStd	6.8	2.2	2.2	2.1	1.8	1.7	1.5	1.8	1.3	2.5	2.3	2.9	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	63.4	72.6	82.0	87.1	93.4	94.8	94.2	86.8	78.7	68.9	61.0		
		MCWB	50.1	50.0	54.8	61.0	68.6	72.7	72.2	72.0	68.3	63.5	54.3	50.6		
	2%	DB	53.2	58.2	68.4	77.1	82.6	89.1	91.3	90.2	82.9	73.9	64.2	53.6		
		MCWB	47.2	47.6	53.3	59.5	65.1	70.7	71.7	71.6	67.6	61.7	53.8	48.2		
	5%	DB	48.6	54.1	64.4	73.2	78.7	85.7	88.2	86.8	79.0	70.4	59.2	49.5		
		MCWB	44.6	45.4	52.0	57.6	63.1	69.2	70.6	70.4	65.2	60.3	52.4	45.4		
	10%	DB	45.6	49.8	60.2	69.4	74.4	82.1	84.7	82.9	75.5	66.6	55.4	46.2		
		MCWB	42.7	43.9	50.0	55.9	61.4	67.7	69.3	68.8	64.0	58.8	51.1	43.1		
	0.4%	WB	51.2	51.0	56.4	62.1	69.4	75.5	75.5	75.0	71.4	65.2	57.3	51.8		
		MCDB	59.1	60.7	68.4	77.6	84.6	89.0	90.8	89.4	81.9	74.1	62.6	57.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.6	48.3	54.3	60.2	66.6	72.5	73.2	72.9	68.4	63.0	55.1	49.0		
		MCDB	51.8	56.2	65.4	75.0	80.1	85.9	88.0	86.9	78.7	70.7	60.9	53.1		
	5%	WB	45.2	46.4	52.5	58.5	64.6	70.4	71.6	71.0	66.6	61.3	53.2	46.1		
		MCDB	47.9	52.5	62.4	71.3	75.8	82.7	85.9	83.9	75.7	68.2	58.5	49.0		
	10%	WB	43.0	44.3	50.7	56.6	62.3	68.6	69.9	69.4	65.0	59.6	51.2	43.8		
		MCDB	45.3	49.3	58.8	67.4	71.9	79.9	82.9	80.8	73.4	65.4	56.0	46.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	16.1	20.9	22.5	21.0	22.2	22.7	22.5	21.1	19.2	15.6	13.3		
		MCDBR	17.6	24.8	29.5	30.7	28.4	27.8	28.1	29.0	26.6	24.7	21.4	18.3		
		MCWBR	12.1	15.0	15.8	14.3	12.6	11.9	11.9	12.4	12.4	13.2	13.0	12.7		
		MCDBR	16.8	22.4	26.3	28.6	25.9	25.8	26.4	26.8	24.1	22.2	18.9	16.9		
	5% WB	MCWBR	12.0	14.2	14.2	13.8	12.1	11.6	11.6	12.1	11.5	12.0	11.8	12.1		
		MCWBR	12.0	14.2	14.2	13.8	12.1	11.6	11.6	12.1	11.5	12.0	11.8	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.324	0.368	0.416	0.427	0.426	0.418	0.408	0.391	0.366	0.329	0.297		
	taud		2.414	2.336	2.273	2.182	2.215	2.265	2.290	2.326	2.358	2.419	2.452	2.442		
	Ebn at Noon		259	271	272	267	268	268	269	267	262	252	244	246		
	Edh at Noon		23	30	37	44	44	42	40	38	34	28	22	20		
(44) All-Sky Solar Radiation	RadAvg		477	754	1138	1456	1695	2017	2028	1757	1344	858	504	403		
	RadStd		55	95	134	168	151	142	153	110	103	92	68	57		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page